Prayas JEE AIR (2025)

Advanced

Weekly Test 26

DURATION : 180 MINUTES | **DATE : 04/01/2025** | **M. MARKS : 180**

ANSWER KEY

PHYSICS

1. (2.00)
2. (3.00)
3. (1.40)
4. (121.57)
5. (0.25)
6. (0.75)
7. (1.00)
8. (2.51)
9. (A, B, C, D)
10. (A, B, C)
11. (B, D)
12. (A, B, C, D)
13. (A, C, D)
14. (B)
15. (B)
16. (D)
17. (B)
18. (B)

CHEMISTRY

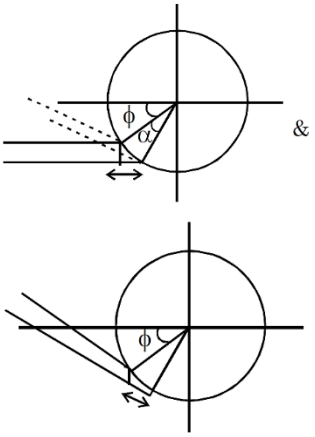
19. (2.25)
20. (4.00)
21. (4.00)
22. (5.00)
23. (5.00)
24. (500.00)
25. (345.00)
26. (6.00)
27. (B, D)
28. (A, B, C)
29. (A, B)
30. (B, C, D)
31. (A, B, C, D)
32. (A, C)
33. (A)
34. (A)
35. (A)
36. (B)

MATHEMATICS

37. (25.00)
38. (26.00)
39. (79.00)
40. (7.00)
41. (273.00)
42. (9.00)
43. (2.00)
44. (0.00)
45. (A, C, D)
46. (A)
47. (A, B, C)
48. (A, B)
49. (A, C)
50. (A, D)
51. (A)
52. (B)
53. (B)
54. (A)

PART-I (PHYSICS)

1. (2.00)

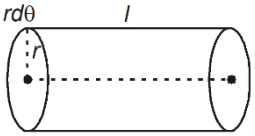


$$\left| r\alpha\sin\phi - r\alpha\cos(\pi-\theta-\frac{\pi}{2}-\phi)\right| = \lambda$$

$$\sin\phi - \sin(\theta+\phi) = \frac{\lambda}{r\alpha}$$

$$r\alpha = \frac{\lambda}{|\sin\phi - \sin(\theta+\phi)|}$$

2. (3.00)



$$dI = \frac{I_0}{\pi R^2}\times(rd\theta)\,dr,\ B = \frac{\mu_0 Ir}{2\pi R^2}$$

$$\therefore\ dF = (dI)\times l\times B$$

$$\therefore\ dP = -\frac{dI\times l\times B}{(l\times rd\theta)}$$

$$\int_P^0 dP = \int_{r=r}^{R} -\frac{I_0}{\pi R^2}\ \frac{(rd\theta)\times(dr)\times l\times\mu_0 Ir}{(l\times rd\theta)\times(2\pi R^2)}$$

$$\Rightarrow P = \frac{\mu_0 I_0^2}{4\pi^2 R^4}\ (R^2 - r^2)$$

$$\therefore \text{At } r = \frac{R}{2},\ P = \frac{3\mu_0 I_0^2}{16\pi^2 R^2}$$

3. (1.40)

Shift, $\Delta x = \left[(\mu-1)t\right]\frac{D}{d}$

$$\Rightarrow 20\beta = (\mu-1)\frac{tD}{d}$$

$$\Rightarrow 20\left(\frac{\lambda D}{d}\right) = (\mu-1)\frac{tD}{d}$$

$$\Rightarrow 20\lambda = (\mu-1)t$$

$$\Rightarrow \mu = \frac{20\lambda}{t}+1$$

$$\Rightarrow \mu = \frac{20\times5000\times10^{-10}}{2.5\times10^{-5}}+1$$

$$\Rightarrow \mu = \frac{4}{10}+1$$

$$\Rightarrow \mu = 1.4$$

4. (121.57)

Energy conservation gives

Energy of H atom in nth state $+\frac{hc}{\lambda_1} = 12.2eV$

$$\Rightarrow\ -\frac{13.6}{n^2}eV + \frac{hc}{\lambda_1} = 12.2eV$$

$$\Rightarrow\ -\frac{13.6}{n^2} + \frac{6.62\times10^{-34}\times3\times10^8}{796\times10^{-10}\times1.6\times10^{-19}} = 12.2$$

$$\Rightarrow\ \frac{13.6}{n^2} = 15.6 - 12.2 = 3.4eV$$

$\therefore\ n = 2$

Hydrogen atom formed is in 1st excited state.

$$\therefore\ \frac{hc}{\lambda_2} = 10.2eV\ \ ;$$

$$\lambda_2 = \frac{1240}{10.2} = 121.568$$

$$\lambda_2 = 121.57\,\text{nm}$$

5. (0.25)

$$dN = \frac{N}{\left(\frac{\pi R}{2}\right)}\left(Rd\theta\right)$$

$$B = \int_0^{\frac{\pi}{2}} \frac{\mu_0 i 2N\times d\theta\times R^2\sin^2\theta}{\pi\times 2R^3} = \frac{\mu_0 iN}{4R}$$

6. (0.75)

In the figure, $\theta = 30°$

When displacement is each ball x, the extension in each spring is $x' = x\cos\theta = x\frac{\sqrt{3}}{2}$

Each spring is connected to two balls.

So, net extension of each spring = $2x' = \sqrt{3}x$

Force along the direction of displacement on each ball is

$$F = k\sqrt{3}x\cos30° + k\sqrt{3}x\cos30° = 3kx$$

$F = -3\,kx$ [negative because restoring in nature]

$$\Rightarrow\quad a = \frac{F}{m} = -\frac{3k}{m}x \Rightarrow \omega = \sqrt{\frac{3k}{m}}$$

$$\Rightarrow\quad f = \frac{1}{2\pi}\sqrt{\frac{3k}{m}} = \frac{1}{\pi}\sqrt{\frac{3k}{4m}} \Rightarrow \beta = \frac{3}{4}$$

$$\beta = 0.75$$

7. (1.00)

$$\omega_1 h\tan\alpha = \omega_2 R \qquad \Rightarrow \frac{\omega_1}{\omega_2} = \frac{R/h}{\tan\alpha} = 1$$

8. **(2.51)**

The instrument has a negative error,

$e = (-5 \times 0.01)$ cm $= -0.05$ cm

Measured reading $= (2.4 + 6 \times 0.01) = 2.46$ cm

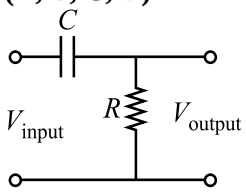
True reading = Measured reading $- e$

$\Rightarrow$ True Reading $= 2.46 - (-0.05)$

$\Rightarrow$ True Reading $= 2.51$ cm

Therefore, diameter of the sphere is 2.51 cm

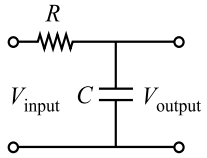
9. **(A, B, C, D)**



(A) $V_{out} = V_{in}\dfrac{R}{z}$

$$V_{out} = \frac{v_0 R}{z}\sin\omega\tau$$

$$v_M = \frac{v_0 R}{z};\ z = \sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}$$

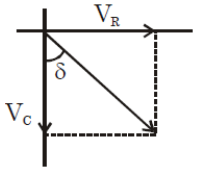


(B) $v_{out} = v_{in}\dfrac{X_c}{z}$

$$v_{out} = \frac{X_c v_0}{z}\sin\omega t, \qquad v_M = \frac{X_c v_0}{z}$$

(C) $P_{av} = \dfrac{v_0 I_0}{2}\cos\phi$, $\qquad \dfrac{v_2^0 R}{2}\times\dfrac{R}{z}$

(D)



$$\tan\delta = \frac{R}{x_C}$$

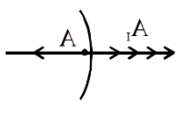
$\tan\delta = \omega RC$

10. **(A, B, C)**

$A \Rightarrow OD = -ut$

$ID = ?$

$$\frac{1}{-ut} + \frac{1}{ID} = \frac{1}{-f}$$

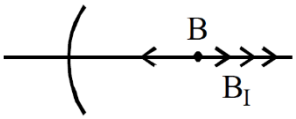


$$\frac{1}{ID} = \frac{1}{ut} - \frac{1}{f}$$

$$ID = \frac{utf}{f - ut} = \frac{f}{\frac{f}{ut} - 1} = +f$$

$B \Rightarrow OD = -(2f - ut)$

$ID = ?$



$$\frac{1}{-(2f - ut)} + \frac{1}{ID} = \frac{1}{-f}$$

$$\frac{1}{\text{ID}} = \frac{1}{2f - ut} - \frac{1}{f}$$

$$\text{ID} = \frac{(2f - ut)f}{-f + ut} = \frac{(2f - ut)f}{-f + ut}$$

$$|ID| = \frac{(2f - ut)f}{f - ut} = f\left(1 + \frac{f}{f - ut}\right) = f\left(1 + \frac{1}{1 - \frac{ut}{f}}\right) = 3f$$

11. **(B, D)**

From the figure it can be seen clearly that energy is released in a nuclear reaction when the B/A of elements produced is more than that of the reactants or the nuclei present before the reaction. If two nuclei in the range $51 < A < 100$ will fuse then they will produce an element with mass number above 100 and less than 200 which has more B/A thus energy is released hence option (B) is correct. Similarly a nucleus in the range $200 <$ A < 260 when broken into two equal fragments then the B/A of these fragments will be more than that of the nucleus hence option (D) is correct.

12. **(A, B, C, D)**

$\vec{F} = -3\hat{i} - 4\hat{j},\ W_{\text{con.f}} = U_i - U_f = 34 - 9 = 25$

& $\dfrac{1}{2}mv^2 - 16 = 34$

13. **(A, C, D)**

$$H = \frac{KAd\theta}{dr} = K\ \ 2\pi rL\ \ \frac{d\theta}{dr}$$

$$\therefore H\int_{r_1}^{r_2}\frac{dr}{r} = 2\pi KL\ \ \theta_2 - \theta_1\ \ = 80\pi$$

$$\Rightarrow \frac{dm}{dt}L = 80\pi$$

$$\Rightarrow \frac{dm}{dt} = \frac{8\pi}{L} = \frac{80\pi}{80\times 4200} = \frac{\pi}{4200}\ \text{Kg/second}$$

14. **(B)**

By law of conservation of linear momentum, we have

$$mv\hat{i} + mv\hat{j} + mv\left(-\hat{j}\right) + \bar{0} = \bar{p}_{\text{TriangularWedgeABC}} + \bar{0}$$

$$\Rightarrow \quad \bar{p}_{\text{TriangularWedgeABC}} = (mv)\hat{i}$$

Since the net linear momentum imparted to the triangular wedge is along *x*-axis and is non-zero, so the centre of mass of wedge *ABC* will move along *x*-axis.

15. (B)

(1) Charge decays exponentially
(2) Charge on capacitor oscillates
(3) Current through inductor oscillates
(4) Current increases exponentially to reach a steady volve.

16. (D)

Use basic concepts

$\Rightarrow$ $Q = CV \Rightarrow$ Ceff in series = $\dfrac{C_1C_2}{C_1+C_2}$

$\Rightarrow$ $E = \dfrac{1}{2}CV^2$

$\Rightarrow$ *PD* in series is distributed as $\dfrac{1}{C_1} : \dfrac{1}{C_2}$

$\Rightarrow$ $C = \dfrac{\in_0 A}{d}$

17. (B)

(I) As velocity = 0 $\Rightarrow$ no radial acceleration

$\Rightarrow T = mg\cos 53° = \dfrac{3mg}{5}$

(II) Motion in horizontal plane $\Rightarrow$ no acceleration in vertical direction

$\Rightarrow T\cos 37° = mg \Rightarrow T = \dfrac{mg \times 5}{4}$

(III) Acceleration of particle w.r.t. cart is zero

$\Rightarrow T\cos 53° = mg \Rightarrow T = \dfrac{5mg}{3}$

(IV) $T = mg\cos 37° \Rightarrow T = \dfrac{4mg}{5}$

18. (B)

(I) $F_v = ma'$

$\rho av^2 = \rho AHa'$

$a' = \dfrac{a}{A}\dfrac{2gh_1}{H}$

$a' = \dfrac{1}{2}\text{m/s}^2$

(II) $a' = \dfrac{a}{A}\dfrac{2gh_2}{H} = 3/2 \text{ m/s}^2$

(III) $\rho av_1^2 + \rho av_2^2 = \rho AHa'$

$\dfrac{a}{A}2g\dfrac{h_1}{H} + \dfrac{a}{A}2g\dfrac{h_2}{H} = a'$

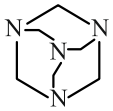
$a' = 2 \text{ m/s}^2$

(IV) Range = $2\sqrt{h(H-h)}$

$\dfrac{R_1}{R_2} = 1$

PART-II (CHEMISTRY)

19. (2.25)


Urotopin

x = 3, y = 6, z = 4

20. (4.00)

Graphite, $(BN)_x$, $B_3N_3H_6$, C_6H_6 have hexagonal planar rings in their structure.

21. (4.00)

$A_n(g) \rightleftharpoons n\,A(g)$

100 0

48 52n

$M\,mix = \dfrac{48}{48+52n} \times 32n + \dfrac{n \times 52}{48+52n} \times 32$

$= \dfrac{32n \times 100}{48+52n}$

Now, $\dfrac{1}{1.25} = \sqrt{\dfrac{32}{\dfrac{32n \times 100}{48+52n}}}$

On solving we get n = 4

22. (5.00)

Moles of Na_3PO_4 = $\dfrac{8.2}{164}$ = 0.05 moles

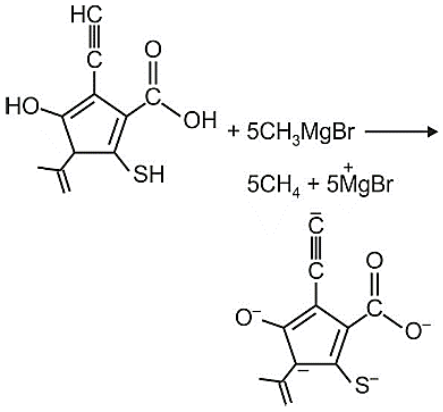
Moles of HCl = $\dfrac{4.5625}{36.5}$ = 0.125 moles

In solution HCl and Na_3PO_4 will react.

Finally solution will have 0.025 mol NaH_2PO_4 and H_3PO_4 (buffer)

$\therefore$ pH = pK_{a_1} + $\log\dfrac{[Salt]}{[Acid]}$ = 5

23. (5.00)



24. **(500.00)**

	A	$\rightarrow 2B$	+ C
t = 0	4a	0	3a
t = t	4a – x	2x	3a + x

From equation (4a – x) (40°) + 2x (10°)+ (3a + x) + (–30°) = 0°

$\therefore x = \frac{7}{5}a$

$$\text{Now, } t = \frac{1}{K}.\ln\frac{4a}{4a-\frac{7}{5}a} = \frac{1}{0.001}.\ln\frac{20}{13} = 500\,\text{min}$$

25. **(345.00)**

Balance eq. is

$$4Au + 8CN^- + O_2 + 2H_2O \rightarrow 4Au(CN)_2^- + 4OH^-$$

(i) $4Au \rightarrow 4Au^+ + 4e$

$\Delta G° = 4\times 96500\times 1.5J$

$4Au^+ + 8CN^- \rightarrow 4Au(CN)_2^-$

$\Delta G° = -2\times 8\times 300\times 38\times 4J$

$4Au + 8CN^- \rightarrow 4Au\,(CN)_2^- + 4e$

$\Delta G_j'^o = 579 - 729.6KJ = -150.6KJ$

(ii) $O_2 + 4H^+ + 4e \rightarrow 2H_2O$

$\Delta G° = -4\times 96500\times 1.2J$

$4H_2O \rightarrow 4H^+ + 4OH^-$

$\Delta G_2'^o = 2\times 8\times 300\times 56$

$O_2 + 2H_2O + 4e \rightarrow 4OH^-$

$\Delta G_2'^o = -463.2 + 268.8kJ = -194.4kJ$

$\Delta G^o_{\text{reaction}} = \Delta G_3^o + \Delta G_3'^o = -345kJ$

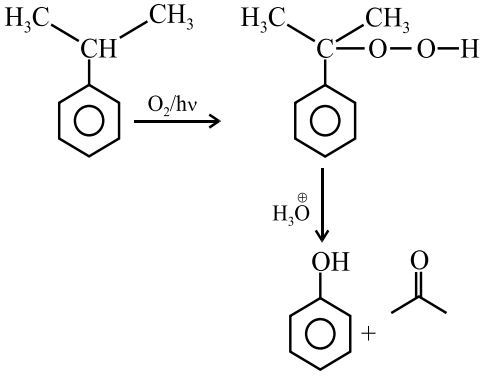
26. **(6.00)**

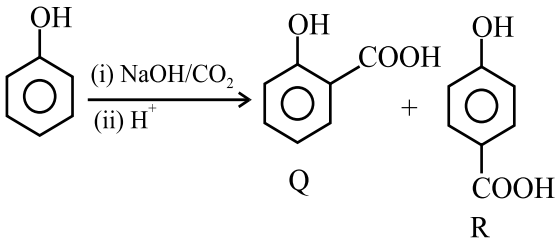
KNO_3, Ag_2O, $KClO_3$, HgO, $NaNO_3$ and H_2O_2 will decompose on heating to give O_2 as the only gaseous product.

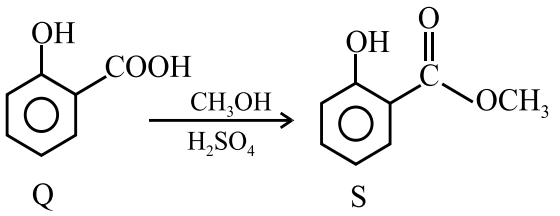
27. **(B, D)**

In dsp^3 hybridisation, inner d_{z^2} orbital is involved dsp^2 hybridisation forms square planar geometry.

28. **(A, B, C)**







29. **(A, B)**

$Cl_2 + H_2O \rightarrow HCl + HOCl$

(x)

$$HCl + AgNO_3 \rightarrow \underset{\text{White}}{AgCl\downarrow} + HNO_3$$

$$2HCl + Mg \rightarrow MgCl_2 + \underset{(y)}{H_2(g)\uparrow}$$

30. **(B, C, D)**

Number of moles of Mg= $\frac{1}{24}$

Number of moles of $O_2 = \frac{1}{64}$

2 Mg	+	$O_2 \rightarrow 2MgO$
Initial moles $\frac{1}{24}$	$\frac{1}{64}$	
Moles at the end $\left(\frac{1}{24}-\frac{1}{32}\right)$	0	$\frac{1}{32}$

Of reaction

Mass of Mg left unreacted

$$= \left[\frac{1}{24}-\frac{1}{32}\right]\times 24 = 0.25g$$

O_2 gas is consumed completely.

Mass of MgO formed $= \frac{1}{32}\times 40 = 1.25g$

31. **(A, B, C, D)**

$XeF_6 + SiO_2 \rightarrow XeOF_4 + SiF_4$

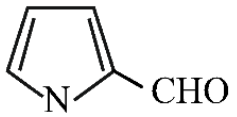
$XeOF_4 + H_2 \xrightarrow{300°C} Xe + H_2O + HF$

$XeO_2F_2 + H_2O \longrightarrow XeO_3 + HF$

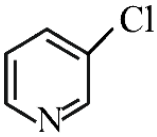
XeO_3 disproportionate in strongly basic medium

32. (A, C)

Reimer-Tiemann reaction.



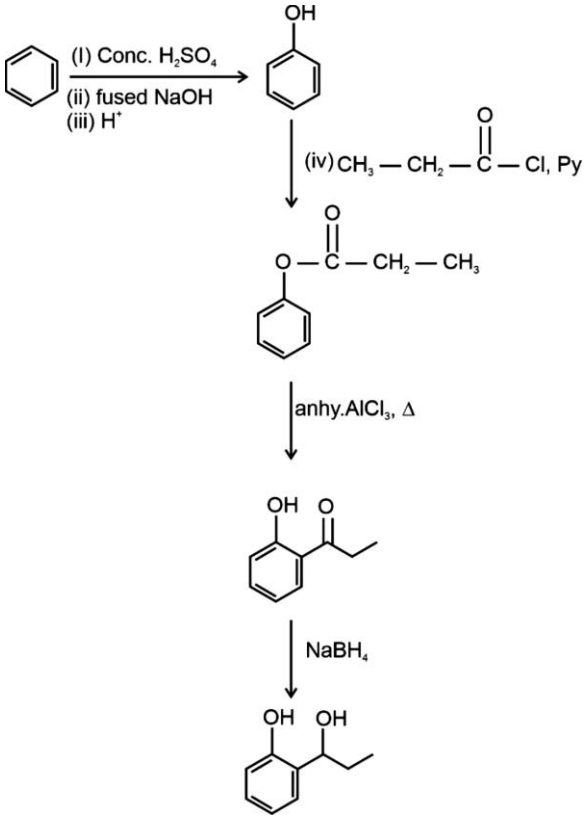
Normal Product



Abnormal RTR product

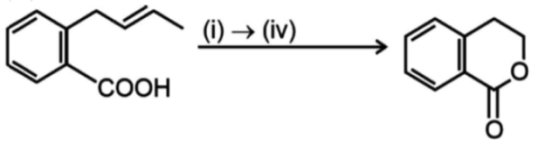
33. (A)

(I)



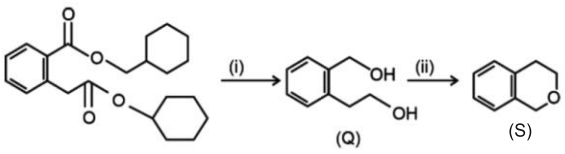
Hence (I → P)

(II)



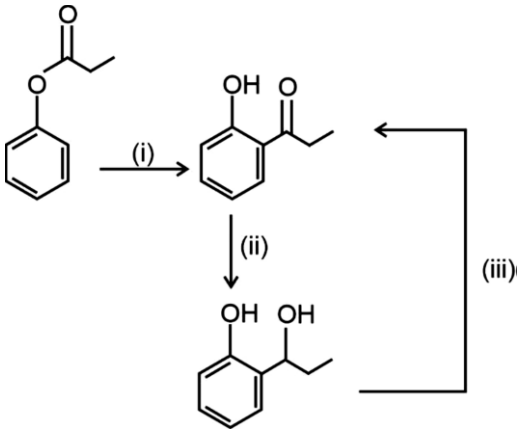
Hence (II →T)

(III)



Hence (III →R)

(IV)



Hence (IV → S)

34. (A)

(P) For isothermal reversible compression of ideal gas, $\Delta U = 0$, $\Delta H = 0$, $\Delta S_{Total} = 0$, $\Delta S_{sys} < 0$, $W \neq 0$ as $\Delta V \neq 0$

(Q) For adiabatic reversible compression of an ideal gas, $q = 0$, $\Delta S_{sys} = 0$ $\Delta S_{unir} = 0$, $\Delta S_{surr} = 0$, $\Delta U \neq 0$, $\Delta H \neq 0$, $W \neq 0$

(R) For adiabatic irreversible expansion $q = 0$, $\Delta U \neq 0$, $\Delta H \neq 0$ $W \neq 0$, $\Delta S_{sys} > 0$

(S) For vaporisation of liquid at normal boiling point, $S_{sys} = \dfrac{\Delta H_{vap}}{T_b}$

Hence, $\Delta S_{sys} > 0$

35. (A)

$\therefore$ b and d are meso

c and d are diastereoisomers

36. (B)

(A) $[Co(en)_3]^{3+}$; Co^{3+} has d^6 configuration which has higher CFSE; so hybridisation is d^2sp^3 and complex is diamagnetic. 'en' is a bidentate chelate ligand.

(B) $[Co(ox)_3]^{3-}$; Co^{3+} has d^6 configuration which has higher CFSE; so hybridisation is d^2sp^3 and complex is diamagnetic. 'ox' is a bidentate chelate ligand.

(C) $[Co(H_2O)_6]^{2+}$; Co^{2+} has d^7 configuration; H_2O is a weak field monodentated ligand, so it is paramagnetic with three unpaired electrons having sp^3d^2 hybridisation.

(D) $[Co(NO_2)_6]^{3-}$; Co^{3+} has d^6 configuration which has higher CFSE; so hybridisation is d^2sp^3 and complex is diamagnetic. 'NO_2^-' is a monodentate ligand.

PART-III (MATHEMATICS)

37. (25.00)

$9 + 9\tan^2 x + 4 + 4\cot^2 x = 13 + (9\tan^2 x + 4\cot^2 x)$

$$\frac{9\tan^2 x + 4\cot^2 x}{2} \geq \sqrt{36} \Rightarrow 9\tan^2 x + 4\cot^2 x \geq 12$$

$$\therefore (9\sec^2 x + 4\operatorname{cosec}^2 x)_{\min} = 13 + 12 = 25$$

38. (26.00)

Since, $\vec{x}, \vec{y}$ and $\vec{x} \times \vec{y}$ are 3 non-coplanar vectors

Hence, $(a-3)\alpha^2 + (b-4)\alpha + (-1) = 0$

$(a-3)\beta^2 + (b-4)\beta + (c-1) = 0$

$(a-3)\gamma^2 + (b-4)\gamma + (c-1) = 0$

We can say that $(a-3)x^2 + (b-4)x + (c-1) = 0$ has roots α, β, γ

Since, quadratic has 3 roots, so it becomes an identity.

$\Rightarrow a-3=0, b-4=0, c-1=0$

$\Rightarrow a=3, b=4, c=1$

$\Rightarrow a^2 + b^2 + c^2 = 26$

39. (79.00)

Let *a*, *b*, *c*, and *d* be the results of rolling the four dice respectively.

We have the range $1 \leq a \leq b \leq c \leq d \leq 6$, and there are $6^4 = 1296$ total outcomes from rolling the dice. To transfer the inequality into a strictly increasing inequality, we can transform it into

$d' = d + 3$. Then, $1 < a' < b' < c' < d' \leq 9$.

Clearly, there are ${}^9C_4 = 126$ values that satisfy this equation, so our answer is $\frac{{}^9C_4}{6^4} = \frac{7}{72} \Rightarrow 079.$

40. (7.00)

$A^2 = 3A + \alpha I$

$A^3 = 3A^2 + \alpha A$

$A^3 = 3(3A + \alpha I) + \alpha A$

$A^3 = 9A + \alpha A + 3\alpha I$

$A^4 = (9+\alpha)A^2 + 3\alpha A$

$= (9+\alpha)(3A + \alpha I) + 3\alpha A$

$= A(27 + 6\alpha) + \alpha(9 + \alpha)$

$\Rightarrow 27 + 6\alpha = 21 \Rightarrow \alpha = -1$

$\Rightarrow \beta = \alpha(9+\alpha) = -8$

41. (273.00)

$5f(x+y) = f(x) \cdot f(y)$

$5f(0) = f(0)^2 \Rightarrow f(0) = 5$

$5f(x+1) = f(x) \cdot f(1)$

$$\Rightarrow \frac{f(x+1)}{f(y)} = \frac{f(1)}{5}$$

$$\Rightarrow \frac{f(1)}{f(0)} \cdot \frac{f(2)}{f(1)} \cdot \frac{f(3)}{f(2)} = \left(\frac{f(1)}{5}\right)^3$$

$$\Rightarrow \frac{320}{5} = \frac{(f(1))^3}{5^3} \Rightarrow f(1) = 20$$

$\therefore \quad 5f(x+1) = 20 \cdot f(x) \Rightarrow f(x+1) = 4f(x)$

$$\sum_{n=0}^{5} f(n) = 5 + 5.4 + 5.4^2 + 5.4^3 + 5.4^4 + 5.4^5$$

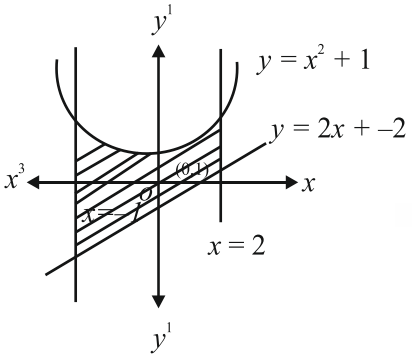
$$= \frac{5\left[4^6 - 1\right]}{3} = 6825$$

42. (9.00)

$$= \int_{-1}^{2} \left[\left(x^2+1\right) - \left(2x-2\right)\right] dx$$

$$\text{Area} = \int_{-1}^{2} \left(x^2 - 2x + 3\right) dx = \left[\frac{x^3}{3} - x^2 + 3x\right]_{-1}^{2}$$

$$= \frac{8}{3} - 4 + 6 - \left(-\frac{1}{3} - 1 - 3\right) = 9$$



43. (2.00)

$$I = \int_1^2 \{x\}\, dx + \int_2^3 \{x\}^2\, dx + \int_3^4 \{x\}^3 dx$$

$$= \int_0^1 x + x^2 + x^3 = \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{6+4+3}{12} = \frac{13}{12}$$

44. (0.00)

$\left(\cos^{-1} x \in [0, \pi]\right)$ but $\sin^{-1}\left(\frac{-1}{4}\right) < 0$

45. (A, C, D)

$I + f = \left(9 + \sqrt{80}\right)^n$

$f' = \left(9 - \sqrt{80}\right)^n$

$I + f + f'$ = even

46. (A)

$g'(x) = (2x^2 - \ln x)\, f(x)$

$f'(x) = \frac{1}{\ln x^3} 3x^2 - \frac{1}{\ln x^2} 2x$

$f'(x) = \frac{x(x-1)}{\ln x} > 0 \ \forall x > 1$; $f(x) > f(1)$

$\Rightarrow f(x) > 0 \ \forall \ x > 1$

For g(x) is increasing

$g'(x) > 0 \Rightarrow 2x^2 - \ln x > 0$ as $(f(x) > 0)$

Let $H(x) = 2x^2 - \ln x$

$H'(x) = 4x - \frac{1}{x} = \frac{4x^2 - 1}{x} > 0$ where $x > 1$

$H(x) > H(1) \Rightarrow (x) > 2$

$\therefore g'(x) > 0 \ \forall \ x \in (1, \infty)$

$\therefore$ g(x) is increasing on $(1, \infty)$

47. (A, B, C)

Common tangents are $y = \pm \frac{x}{\sqrt{3}} \mp \sqrt{3}$

48. (A, B)

$\sin^2 x + \sin x = (a + 2)$

$\therefore y = \sin^2 x + \sin x = \left(\sin x + \frac{1}{2}\right)^2 - \frac{1}{4}$

$\Rightarrow y \in \left[\frac{-1}{4}, 2\right]$

So, $\frac{-1}{4} \le a + 2 \le 2 \Rightarrow \frac{-9}{4} \le a \le 0$ if $a \in I$

$\Rightarrow a = -2, -1, 0$

49. (A, C)

$a + b + c + d + e + f = 30$

find solutions per constraints

50. (A, D)

Domain of $f(x)$ is $(-1,1)$. So, $\left[x^2\right] = 0$

51. (A)

(I) Tangent at the each of focal chord meet at 90° and meet at directrix

(II) The angle between *SP* and *SQ* is 90°

(III) Condition for focal chord $t_1 t_2 = -1$

(IV) Condition for normal chord

52. (B)

(I) $\frac{dy}{dx} = e^x (A\cos x + B\sin x) + e^x(-A\sin x - B\sin x)$

$= y + e^x(-A\sin x + B\cos x)$

$\therefore \quad \frac{d^2y}{dx^2} = \frac{dy}{dx} + e^x (-A\sin x + B\cos x) + e^x(-A\cos x - B\sin x)$

$= \frac{dy}{dx} + \left(\frac{dy}{dx} - y\right) - y = 2\left(\frac{dy}{dx} - y\right)$

So, degree = 1 and order = 2

(II) The equation of the family is $y^2 = 4a(x - b)$

Where a, b are arbitrary constants

$\therefore \quad 2y\frac{dy}{dx} = 4a \quad$ or $\quad \left(\frac{dy}{dx}\right)^2 + y\frac{d^2y}{dx^2} = 0$

So, degree = 1 and order = 2

(III) The equation of the family is $(x - c)^2 + y^2 = c^2$

or $x^2 + y^2 - 2cx = 0$ or $\frac{x^2 + y^2}{x} = 2c$

$\Rightarrow \quad \frac{\left(2x + 2y\frac{dy}{dx}\right)x - (x^2 + y^2)\cdot 1}{x^2} = 0$

So, degree = 1 and order = 1

(IV) The equation of the family is

$\frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} = 1$ because they have the same foci $(\pm\sqrt{a^2 - b^2}, 0)$

On differentiating, $\frac{2x}{a^2 + \lambda} + \frac{2y}{b^2 + \lambda} \cdot \frac{dy}{dx} = 0$

or $\frac{x}{a^2 + \lambda} + \frac{yp}{b^2 + \lambda} = 0 \quad \left(\text{Let } p = \frac{dy}{dx}\right)$

or $x(b^2 + \lambda) + yp(a^2 + \lambda) = 0$

$\Rightarrow \quad \lambda = \frac{-b^2x - a^2yp}{x + yp}$

$\therefore$ The differential equation is

$$\frac{x^2}{a^2 + \frac{-b^2x - a^2yp}{x + yp}} + \frac{y^2}{b^2 + \frac{-b^2x - a^2yp}{x + yp}}$$

or $\frac{x(x + yp)}{a^2 - b^2} + \frac{y(x + yp)}{(b^2 - a^2)p} = 1$

So, Order = 1 and degree = 2

53. (B)

Given $\frac{abc}{6} = 32$, where *A*, *B*, *C* are respectively $(a, 0, 0), (0, b, 0), (0, 0, c)$

(I) Centroid of tetrahedron $(\alpha, \beta, \gamma) = \left(\frac{a}{2}, \frac{b}{2}, \frac{c}{2}\right)$

$\Rightarrow a = 4\alpha, b = 4\beta, c = 4\gamma$

$\therefore 64\,\alpha\beta\gamma = 32 \times 6 \Rightarrow \alpha\beta\gamma = 3$

(II) Equidistant point $(\alpha, \beta, \gamma) = \left(\frac{a}{2}, \frac{b}{2}, \frac{c}{2}\right)$

$\Rightarrow a = 2\alpha, b = 2\beta, c = 2\gamma$

$\therefore 8\alpha\beta\gamma = 32 \times 6 \Rightarrow \alpha\beta\gamma = 24$

(III) C is (0, 0, 8)

(IV) D is (0, 4, 0)

54. (A)

i. $z = w$ or w^2

ii. $z = \pm i$

iii. $z = \pm 1$



PW Web/App - https://smart.link/7wwosivoicgd4

Library- https://smart.link/sdfez8ejd80if